

APEN 2026 Competition Handbook

APEN & Imperial EdTech

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CONTENTS

APEN AI, Coding & Robotics Competition Handbook	
1. Welcome	
2. Competition Overview	
3. AI in the Competition	
Generative AI & Vibe Coding	
Modalities	
4. Equipment & Resources	
5. Competition Timeline — 9-Week Cycle	
6. Stages of the Competition	
7. Division Niche Breakdown	
Primary School Division — Ages 7–12 — Agriculture	
Secondary School Division — Ages 13–16 — Power	
Sixth Form Division — Ages 16–18 — Security	
8. Judging Criteria	
9. Deliverables per Team	
10. Prizes & Awards	
11. How to Participate	
12. Participant Agreement	
13. Conclusion	
14. Contact	

APEN AI, Coding & Robotics Competition Handbook

APEN 2026 · Lagos (Hybrid) · Organized by the Association of Private Educators in Nigeria (APEN), in partnership with Imperial EdTech

Theme: “Smart Solutions for Tomorrow’s Nigeria — Design. Build. Intelligize.”

1. Welcome

Welcome to the APEN 2026 AI, Coding & Robotics Competition. This year sharpens the focus: each age division owns one real-world Nigerian challenge, solved with hardware engineering and generative AI.

- Primary Schools tackle **Agriculture**
- Secondary Schools tackle **Power**
- Sixth Form Colleges tackle **Security**

The competition pushes students to innovate at the intersection of physical computing and AI. The 2026 edition introduces a 4-stage Innovation Funnel that takes teams from pure design thinking, through a standardized hardware build, into open-ended AI integration (“Intelligize”), and culminates in a live Grand Finale Battle. Every team receives the same base kit; the AI layer is left open to their creativity — fair judging, genuine innovation rewarded.

2. Competition Overview

All projects follow a structured Engineering Design Thinking framework:

1. **Empathize** — Conduct field research. Identify real pain points in Nigerian farms, homes, or communities.
2. **Define** — Frame the challenge from that research. Set measurable goals.
3. **Ideate** — Sketch design concepts. Select the most feasible hardware and AI approach.

4. **Prototype** — Assemble the standardized APEN kit. Run real-world tests.
5. **Improve** — Integrate the AI layer. Fix flaws found in testing. Optimize the experience.
6. **Communicate** — Present the solution, its impact, and what the AI layer adds.

3. AI in the Competition

GENERATIVE AI & VIBE CODING

Coding is evolving. Foundational logic still matters, but students now use AI to accelerate development:

- **Prompt Engineering** — directing large language models to interpret sensor data, generate alerts, or write logic blocks.
- **Vibe Coding** — building dashboards and data-logging interfaces by describing intent to an AI rather than writing syntax from scratch.
- **Edge AI & Computer Vision** — advanced teams may run lightweight ML models directly on microcontrollers to recognize patterns or anomalies.

MODALITIES

- **Physical Hands-on** — direct work with the APEN kit and add-ons.
- **Virtual Online** — remote AI integration, dashboard building, digital submissions.
- **Hybrid Learning** — physical builds combined with virtual AI mentorship.

4. Equipment & Resources

Each division uses one common kit tailored to its niche, so every team in a division builds the same core project and judging stays fair and consistent. An **Innovation Store** is available to all participating schools — because Stage 3 is open-ended, teams may purchase add-on component packs (extra sensors, camera modules, connectivity boards, power add-ons) to push their AI layer further.

Safety note (Secondary — Power Division): all power-sensing activities use a low-voltage DC proxy circuit to represent grid power. No team wires directly into live 220V household current.

5. Competition Timeline — 9-Week Cycle

Built to fit the 2026/2027 academic calendar, spanning pre-registration through the Grand Finale.

Phase	Dates	Focus	Deliverable
1 — Pre-registration & kit dispatch	Aug 31 – Sep 30, 2026	Registration closes Sep 25; kits dispatch Sep 14–30	—
2 — Stage 1 Design Sprint	Sep 28 – Oct 4, 2026	Design thinking & problem definition	3-minute video pitch, due Oct 4
3 — Stage 2 Advanced Build	Oct 5 – Oct 23, 2026	Hardware assembly & testing	Functional prototype, due Oct 23
4 — Judging & mid-term break	Oct 24 – Nov 8, 2026	Preliminary judging; feedback integration	Finalists selected
5 — Stage 3 Finals Prep	Nov 9 – Nov 22, 2026	AI integration & Intelligize layer	Final AI prompt log & video, due Nov 22
6 — Grand Finale (BATTLE)	Nov 26, 2026	Live presentations, judging, awards	APEN 2026 Champions crowned

6. Stages of the Competition

Stage 1 — Design (the gateway qualifier). Pure design thinking and empathy. Conduct local field research, interview users, identify real pain points. Deliverable: 3-minute video pitch by October 4, 2026. Teams that show clear problem understanding advance, and their kit is dispatched.

Stage 2 — Build (the standardized hardware phase). Closed-ended technical execution and fairness. Assemble and program the official APEN division kit so core hardware logic functions reliably. Deliverable: functional prototype by October 23, 2026. Every team uses identical hardware, so judging is consistent across all schools.

Stage 3 — Intelligize (the open AI innovation phase). Open-ended AI integration and creativity. Teams integrate generative AI, machine learning, vibe coding, or edge AI to turn raw hardware data into human insight, drawing on the Innovation Store for any additions their vision requires. Deliverable: final AI prompt log and project video by November 22, 2026.

Stage 4 — Battle (the Grand Finale). Top-performing teams from Stages 1–3 are selected as finalists. On November 26, 2026, finalists present a 3–5 minute final pitch covering hardware functionality, AI layer integration, and real-world impact, live in Lagos (virtual and/or physical). Teams are ranked by judges; winners are crowned APEN 2026 Champions.

7. Division Niche Breakdown

PRIMARY SCHOOL DIVISION — AGES 7–12 — AGRICULTURE

Project: Smart Farm Bot + LED Interface **Challenge:** Design and build a Smart Farm Bot that helps a plant grow using less water, less space, and less human effort — then make it intelligent. (SDG 2, SDG 6) **Core hardware:** ESP32-S3 microcontroller, environmental sensors, actuators, LED interface **Kit price: ₦60,050 per team**

Component	Qty	Unit price
ESP32 38-Pin Type-C Development Board	1	₱16,000
USB Type-C Data/Programming Cable	1	₱3,500
DHT11 Temperature & Humidity Sensor	1	₱2,000
LDR/Photoresistor Light Sensor	1	₱1,100
Water-Level Sensor	1	₱3,500
Water-Flow Sensor	1	₱5,500
5V DC Water Pump	1	₱6,600
1-Channel Relay Module	1	₱3,400
Mini DC Fan	1	₱4,500
LED Grow Light	1	₱1,000
Active Buzzer	1	₱1,450
Red / Green / Yellow-Blue LED	3	₱3,300
400-Hole Mini Breadboard	1	₱3,200
Packaging & labelling	1	₱5,000

SECONDARY SCHOOL DIVISION — AGES 13–16 — POWER

Project: Smart Energy Bot **Challenge:** Use technology to monitor electricity consumption, detect energy waste, and use AI to help homes and schools make smarter, more environmentally friendly energy decisions. (SDG 7, 9, 12) **Core hardware:** ESP32 board, voltage and current sensors, actuators **Safety:** low-voltage safe proxy circuit only — no direct 220V mains wiring permitted **Kit price: ₱63,100 per team**

Component	Qty	Unit price
ESP32 Development Board – 38 Pin Type-C	1	₱16,000
ZMPT101B AC Voltage Sensor	1	₱6,500
ACS712 20A Current Sensor	1	₱4,800
16×2 LCD + I2C Module	1	₱7,500
Push Button Tact Switch Module	1	₱2,100
MB-102 830-Point Breadboard	1	₱3,200
400-Hole Mini Breadboard	2	₱5,000
Jumper Wire Set (M-M/M-F/F-F)	1 pack	₱6,000
5V USB Power Supply	1	₱4,500
USB Type-C Cable	1	₱2,500
Packaging	1	₱5,000

SIXTH FORM DIVISION — AGES 16–18 — SECURITY

Project: ESP32-CAM Smart Security Bot **Challenge:** Design an affordable smart security system that detects movement, captures visual evidence, alerts a user remotely, and uses AI to judge whether an observed event matters. (SDG 9, 11, 16)

Core hardware: ESP32-CAM, PIR sensor, camera, Wi-Fi, phone alerts **Kit price:** ₱64,400 per team

Component	Qty	Unit price
ESP32-CAM with OV2640 Camera	1	₦24,000
ESP32-CAM MB USB Programming Board	1	₦5,000
HC-SR501 PIR Motion Sensor	1	₦3,000
830-Point Solderless Breadboard	1	₦4,500
140-Piece Mixed Jumper Wire Set	1 pack	₦7,000
LM2596 DC-DC Buck Converter	1	₦3,800
18650 Li-ion Battery	2	₦8,000
18650 Battery Holder	1	₦2,100
JST/Power Connector Set	1 set	₦2,000
Packaging / student kit case & labelling	1	₦5,000

Logistics: ₦10,000 flat delivery within Lagos, ₦20,000 outside Lagos, added at checkout. Challenges may be adjusted for material availability — alternative components will be provided where needed.

8. Judging Criteria

Scores are aggregated across all four stages of the Innovation Funnel.

Criterion	Weight	What's assessed
Design Thinking & Empathy	20%	User research evidence, problem definition clarity, ideation, SDG alignment
Hardware Execution & Build Quality	30%	Build functionality and reliability, code quality, sensor calibration, actuator triggers
AI Innovation & the Intelligize Layer	30%	Utility and creativity of AI integration, prompt engineering sophistication, vibe coding / edge AI complexity, prompt-log documentation
Presentation & Documentation	20%	Clarity of the solution and journey, technical documentation quality, demonstration confidence, team collaboration

9. Deliverables per Team

Stage 1 Video Pitch (3 min) — due October 4, 2026 Field research and pain points, the proposed kit solution and real-world application, SDG alignment, and how the solution will be built and tested.

Physical Hardware Prototype — due October 23, 2026 A fully functional model built strictly from the official APEN division kit, demonstrating reliable sensor readings and accurate actuator triggers.

Final Video Pitch (3–5 min) — due November 22, 2026 The physical hardware in action with the AI layer working seamlessly alongside it in real time. Teams advancing to the Grand Finale present this live on November 26, 2026.

AI Prompt Log & Design Poster — due November 22, 2026 A poster summarizing the project journey and circuit design, plus a full log of every LLM prompt, prompt iteration, vibe-coding workflow, and edge-AI implementation used to reach the final build.

10. Prizes & Awards

Cash grants, scholarships, and school materials and resources are awarded across all three divisions, including:

- Advanced robotics kits and tech gadgets
- Cash grants and technology coupons
- Trophies and medals
- Closed-door mentorship sessions with industry leaders in agri-tech, renewable energy, and security technology
- Networking with professionals in innovation and tech entrepreneurship
- Recognition as APEN 2026 Champions

11. How to Participate

Step 1 — Register & pitch. The school's ICT or STEM teacher registers for their division and completes kit payment during pre-registration (Aug 31 – Sep 25, 2026). Join the official announcements channel. The student team records and uploads the Stage 1 video pitch by October 4.

Step 2 — Receive & build. Receive the official APEN kit during the dispatch window (Sep 14 – Sep 30). Join live AI and coding mentorship sessions. Assemble and test the standardized hardware through October 23, ready for judging.

Step 3 — Intelligize & innovate. During Phase 5 (Nov 9 – Nov 22), integrate the chosen AI, ML, or vibe-coded layer, drawing on the Innovation Store for any add-ons. Upload the final video, design poster, and AI prompt log by November 22.

Step 4 — Grand Finale & recognition. Top finalists present live in Lagos (virtual and/or physical) on November 26, taking part in live judging, Q&A, and the award ceremony. Champions are crowned and receive prizes and mentorship.

12. Participant Agreement

All participants agree to:

- Use only approved, official APEN division kit components for the Stage 2 standardized build
- Submit original work and document all AI prompt engineering in the prompt log
- Follow the Engineering Design Thinking process throughout
- Meet all submission and presentation deadlines across all six phases
- Take part in required presentations — Stage 1 video pitch, final video pitch, Grand Finale Battle
- Comply with safety rules: Secondary Power Division teams use only low-voltage DC proxy circuits; no direct 220V mains wiring, ever
- Attend the live Grand Finale Battle presentation on November 26, 2026 if advancing as a finalist, or forfeit the finalist position
- Respect the decisions of the judging panel and industry experts

13. Conclusion

APEN 2026 integrates engineering design thinking with hands-on tech skills, empowering students to solve real Nigerian challenges in agriculture, power, and security. By following a structured problem-solving approach and incorporating AI, participants develop innovative, user-focused solutions for tomorrow's Nigeria.

The Grand Finale Battle on November 26, 2026 showcases the best of these innovations and crowns the APEN 2026 Champions — the result of a rigorous nine-week cycle spanning four stages: Design, Build, Intelligize, and Battle.

14. Contact

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